14.2.6 Structure of proteins

Proteins are polymers of amino acids. Their three dimensional structure depends mainly on the sequence of amino acids (residues). The protein structure can be described at four hierarchal levels called primary, secondary, tertiary and quaternary structures as shown in the figure 14.16

1. Primary structure of proteins:

Proteins are polypeptide chains, made up of amino acids are connected through peptide bonds. The relative arrangement of the amino acids in the polypeptide chain is called the primary structure of the protein. Knowledge of this is essential as even small changes have potential to alter the overall structure and function of a protein.

H_2N—Gly—Met—Phe—Cys—Arg—Asp—COOH

2. Secondary structure of proteins:

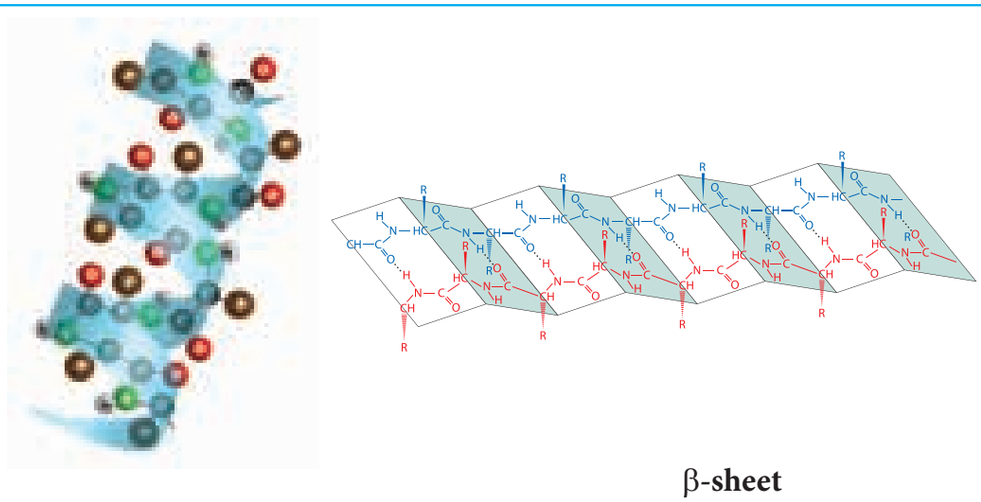
The amino acids in the polypeptide chain forms highly regular shapes (sub-structures) through the hydrogen bond between the carbonyl oxygen (>C=O) and the neighbouring amine hydrogen (-NH)of the main chain.α-Helix and β-strands or sheets are two most common sub-structures formed by proteins.

α-Helix

In the α-helix sub-structure, the amino acids are arranged in a right handed helical (spiral) structure and are stabilised by the hydrogen bond between the carbonyl oxygen of one amino acid (n^{th} residue) with amino hydrogen of the fifth residue ($n+4^{th}$ residue). The side chains of the residues protrude outside of the helix. Each turn of an α-helix contains about 3.6 residues and is about 5.4 Å long. The amino acid proline produces a kink in the helical structure and often called as a helix breaker due to its rigid cyclic structure.

β-Strand

β-Strands are extended peptide chain rather than coiled. The hydrogen bonds occur between main chain carbonyl group one such strand and the amino group of the adjacent strand resulting in the formation of a sheet like structure. This arrangement is called β-sheets.



α-Helix β-sheet

Figure 14.16 Secondary structure of proteins

XII U14-Biomolecules_New.indd 254 11-01-2021 16:37:50